AI UNIVERSAL
MAKING WAVES WITH PHYSICS-DRIVEN AI
For People, Planet, Business

AI UNIVERSAL
10 Rue Dupont des Loges
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Full Stack Engineering Internship
3rd Year Engineering (BAC+5) 6 months
12 months Internship

CANDIDATE SELECTION
TECHNICAL ASSESSMENT

1 Marine Fleet Control Center

Imagine you are the head of a fleet centre responsible for monitoring 1000 of ship operational performance in real-time.

In your fleet, each vessel is equipped with a multi-sensor data acquisition (DAQ) system that continuously measures key physical and operational parameters. This includes GPS, speed, heading, fuel consumption meter and motions sensors.

These measurements are timestamped and transmitted via satellite communication to a shore-based database infrastructure, where they are stored, processed, and made accessible for analysis through an API.

Using such API gives you an overview of fleet states including historical time-series navigation and operational data, using unique identification number called IMO.

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2 FLEET DATA APIs

2.1 FleetView API

FleetView of AIU retrieves all data records onboard ships and stored on the fleet data acquisition database.

These data are timeseries and cover the following ship navigation information:

Parameter	Description	Unit	Notes
Latitude / Longitude	GPS geographic position of the vessel	°	*WGS-84 datum* *Frequency: HF, MF*
SOG	Speed Over Ground — actual speed relative to the seabed	knots	*SOG is not STW (Speed Through Water).* *SOG = STW + Current (vectorial representation)* *Frequency* available: HF, MF*
GPS Course	True direction of travel over the ground	°	*Influenced by currents and wind drift* *Frequency available: HF, MF*
Gyrocompass (Heading)	Direction the vessel's bow is pointing	°	*Differs from course in the presence of currents* *Frequency available: HF, MF*
Thruster Torque	The mechanical torque transmitted from the main engine through the main shaft and gearboxes to drive the thrusters	N * meter	*Frequency* available: HF*
Thruster RPM	Rotations per minute of the main propulsion system	RPM	*Simple model: RPM = 4 × SOG* *Frequency available: HF, MF*
IMU (6-DOF)	Roll, pitch, yaw, heave, surge, sway from inertial sensors	Various	*Frequency available: UH For HF*
Fuel Flow Rate	Instantaneous fuel consumption No available but can be computed	t/day	*When not available, consider model: consumption* $\propto SOG^3$ *At 15 knot speed ship consumes ≈ 150 t per day)* *1 tones of fuel = 1000 USD* *Frequency available: HF, MF, LF*
Onboard systems collect data at three frequency levels: - High Frequency (HF) — sub-second sampling (e.g., inertial motion data) - Medium Frequency (MF) — one sample per 15 minutes (primary dataset for this exercise) - Low Frequency (LF) — one sample per 24 hours (daily summaries, fuel accounting)			

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2.2 AIS API (Alternative to FleetView)

The Automatic Identification System (AIS) is a mandatory maritime tracking system that serves as an alternative data source when direct access to onboard sensor data via the FleetView API is unavailable.

The AIS API provides vessel identification (IMO number, MMSI, name, call sign), position (latitude/longitude), navigation data (course over ground, heading), speed (speed over ground), voyage information (destination, navigational status, ETA, when available), vessel characteristics (type, length, beam, and other static data), and a timestamp for each message. The exact fields vary by provider and by what each vessel transmits

AIS-equipped vessels broadcast this data via VHF radio, which is picked up by coastal stations near shore and satellite receivers over open ocean, then aggregated by AIS providers into global vessel-tracking databases accessible via HTTPS/JSON APIs.

The result is a continuous time series of vessel positions and navigation data used to reconstruct trajectories and monitor fleet movements.

GoodWeather® API

Deployed in production, this API takes the vessel's spatio-temporal position as input and returns detailed oceanic and atmospheric conditions for that location and time. The data is provided at high spatial resolution (1/36°,1/12°, 1/10°, ¼° grid depending on the location and forecast ranges) with a forecast horizon of up to 10 days, enabling accurate environmental inputs for vessel performance and routing models.

3 AIU Proprietary APIs

3.1 GoodWeather® API

This API provides global meteocean high-resolution data (1/36°, 1/12°, 1/10°, and ¼° grid resolutions, depending on location and forecast horizon), covering forecast horizons of 5, 10, 15, and up to 35 days. It delivers accurate environmental inputs for vessel performance analysis and weather routing optimisation. ShipTwin API – Predictive Performance Model

3.2 ShipTwin® API

This API is a library of models that captures the relationship between various vessel parameters under specific weather and loading conditions.

The API takes the following inputs:

- Environmental conditions: wind, wave state, currents, and other relevant meteorological and oceanographic data.
- Vessel configuration: draft, trim, loading condition, and other relevant vessel characteristics.
- Navigation parameters: vessel heading on a route segment.
- Propulsion control parameter: Main Engine RPM, the primary operational parameter used to control vessel speed.
- Based on these inputs, the model predicts the vessel's expected performance, including:
- Speed Over Ground (SOG)
- Fuel consumption
- Other relevant performance parameters

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Conversely, for a target SOG, the API library model can determine the required Main Engine RPM needed to achieve that speed under the given vessel and environmental conditions.

This API enables the AIU Copilot® platform to deliver accurate voyage optimisation and operational recommendations based on multiple objectives and constraints.

3.3 BestRoute API – Weather routing algorithms

In a production, this API takes as an input the departure and arrival location and timing, and return optimal route accounting for dynamic weather obstacles and static obstacles such as landmarks and shallow water.

3.4 OnTime API – Just on Time arrival using min fuel / emissions

In a production this API takes as an input a fix route and weather data and return RPM instruction for an on-time arrival using minimum fuel consumption and emissions.

4 Technical Test Description - Full-Stack Engineer Intern

4.1 Part #1

For remote assessments, no API access will be provided. In addition, the scope has been intentionally simplified in order to focus on core analytical and technical skills.

Data Source

The dataset contains records from three vessels:

- IMO1
- IMO2
- IMO3

The dataset contains positional and operational time-series data collected over a ten-month period.

Objective

Develop a **mini web application** that allows a **fleet control manager** to **replay and analyze the historical operations** of any vessel in the fleet.

Core Features

The application must allow the user to:

1. Select one or more vessels from the three available (IMO1, IMO2, IMO3).
2. Define a time window by choosing a start date and an end date.
3. Select one or more spatio-temporal variables to visualise (e.g., Speed over time, RPM over time).
4. Display each selected vessel's trajectory on an interactive world map.
5. Dynamically colour-code the trajectory based on the selected variable (e.g., colour gradient from low to high speed).
6. Any other feature would be welcome

4.2 Part #2

The test will be carried out in premis

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5 Expected Deliverables

#	Deliverable	Description
1	Frontend Application	Design a small web application with a modern, visually compelling interface. Draw inspiration from best-in-class navigation and weather applications (Apple Weather, Google Waze, Windy.com) without needing to replicate them exactly. Focus on clear data presentation and intuitive UX.
2	Database Architecture	Propose a database schema suitable for storing and efficiently querying the vessel time-series data. Justify your technology choices (e.g., PostgreSQL with TimescaleDB, InfluxDB, etc.) and describe the key tables, indexes, and query patterns. How would you deal with the storage and retrieval of Global Weather Data ?
3	Advanced Feature Proposal	Describe how a user could interactively modify a segment of a vessel's trajectory by dragging a waypoint on the map. Detail the technical approach: data model changes, backend logic, and front-end interaction patterns.

6 Evaluation Criteria

Candidates will be assessed across the following dimensions:

Criterion	What we look for
Code quality & architecture	Clean, readable, well-structured code. Logical separation of concerns. Appropriate use of frameworks and libraries.
UI/UX design	Aesthetics, layout clarity, responsiveness, and ease of use. Map integration quality.
Data modelling	Relevance and scalability of the proposed database schema. Understanding of time-series data constraints.
Problem-solving & initiative	How you handle ambiguity, the assumptions you make, and the depth of your advanced feature proposal.

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7 Reference Applications & Data Sources

The following applications demonstrate best-in-class maritime, weather, and navigation data visualisation. Study them for UX inspiration:

- Apple Weather (iOS) — Clean temporal data presentation, progressive disclosure of detail
- Google Waze — Real-time route visualisation, dynamic colour-coded paths, interactive map controls
- Windy.com — Global meteorological data overlaid on an interactive map; excellent reference for data-dense visualisations
- Zoom Earth
- hifleet.com/kdistance/ — Maritime distance calculation tool
- https://www.sofarocean.com/

7.1 Open Meteorological Data (for inspiration only)

Global oceanographic and meteorological datasets are freely available for download and integration into the application:

- Copernicus Marine Service — https://data.marine.copernicus.eu
- Copernicus Climate Data Store — https://cds.climate.copernicus.eu

Note to Candidate

There is no single correct solution to this exercise. We are interested in how you think, design, and justify your choices. Feel free to go beyond the minimum requirements — creativity and initiative are highly valued. Document any assumptions you make and be prepared to discuss your decisions during the interview.

8 Deadline

Deadline is 26 Aug 2026 @ 18:00. If the result is acceptable, the candidate will be invited for in person test 27 or 28 Aug 2026.

9 Contact

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